DeepThink: Multimodal Autonomous Document Assistant

SLIIT Codefest 2026 AI Competition • Powered by IFS

Chosen Sub-track: Sub-track 1A — Rich Answers, Not Just Text	**Team Name:** DeepThink
Target Corpus: The Ashen Era Archive (415 docs, 1,277 pages)	**Submission Date:** September 2026

■ **Unlisted YouTube Demonstration Video:** https://youtu.be/DEEPTHINK_CODEFEST_DEMO
(Features 4-min live unedited end-to-end screen recording answering real multimodal archival queries + system architecture walkthrough)

1. Executive Summary

Enterprise technical documentation inherently mingles scanned schematics, multi-column tables, visual maps, and complex descriptive narratives across thousands of pages. Traditional Retrieval-Augmented Generation (RAG) suffers from **'text-only myopia'**: systems retrieve text snippets, describe diagrams in vague words, and omit authentic data tables. **DeepThink** solves this challenge for **Sub-track 1A (Rich Answers, Not Just Text)** against the competition's official *Ashen Era Archive* (415 documents, ~1,277 pages across PDF, DOCX, Markdown, Text, and Scans).

The Core Innovation: DeepThink establishes an *autonomous multimodal retrieval contract*. When an inquiry pertains to visual artifacts, architectural blueprints, battle maps, or financial ledgers, DeepThink directly retrieves, verifies, and embeds authentic figures and rendered GitHub Markdown tables inline alongside rigorous, citation-backed textual explanations. The system runs on a **100% free, local-first architecture** (zero credit cards, zero proprietary API embeddings) leveraging local ChromaDB with SentenceTransformers (BAAI/bge-small-en-v1.5) and ultra-low-latency Groq Cloud LPU inference (300+ tokens/sec, ~1.35s total latency).

2. Problem Statement & Sub-track 1A Alignment

The Challenge: The Ashen Era Archive is a completely invented fantasy corpus containing four full novels, ~90 wiki articles, 3 codex data books loaded with figure plates and tables, and ~150 scanned ephemera pieces (trial transcripts, handwritten ledgers, letters). Because the world is entirely synthetic, public LLMs have zero pre-training knowledge. The system must work solely from extracted archival evidence.

Why Sub-track 1A? The competition brief notes that *'a strong, working solution to a single sub-track will always beat a shallow attempt at several.'* Our technical investigation revealed that the vast majority of actionable enterprise facts in technical codexes reside inside figure plates and tables. Sub-track 1A requires the assistant to answer questions using this multimodal content and embed diagrams, tables, and images directly into responses so users inspect actual figures rather than textual hallucinations. DeepThink treats multi-hop relational retrieval (1B) and iterative human-like query decomposition (1C) as foundational internal enablers for Sub-track 1A, ensuring that complex comparative figure queries retrieve all necessary context.

3. Solution Overview & Autonomous Compound RAG 3.0 Architecture

DeepThink replaces naive one-shot vector search with an **8-Stage Autonomous Multimodal Pipeline**, operating without manual UI sliders or heuristic human guesswork:

Stage 1: Pre-Retrieval Domain Guardrail • *Fast-path inspection refuses off-domain queries (real-world history, coding, abuse, physical sensor prompts) in <0.005s with 0 tokens spent.*
Stage 2: Entity Matcher & 3-Stage Context Budgeter • *Extracts proper nouns and maps queries to 16 canonical field families. Dynamically sizes retrieval budget: K=2 (single facts/visuals), K=4 (lore), K=6..8 (comparative matrices).*
Stage 3: Agentic Decomposition & Dense Retrieval • *Decomposes complex comparative questions into focused sub-queries. Ingests dense embeddings locally via BAAI/bge-small-en-v1.5 in ChromaDB.*
Stage 4: Cross-Encoder Multi-Modality Re-Ranking • *Applies primary Character Dossier boost (+0.35), Multi-Entity intersection boost (+0.40), and Modality factor (1.5x for images/tables).*
Stage 5: Pre-Generation Verification & Noise Pruning • *Elbow-method prune drops trailing chunks (score < 0.60 × TopScore). Verifies entity presence and enforces 2,500 token budget.*
Stage 6: Grounded Generation (Groq LPU) • *Ultra-fast inference (300+ t/s) with Directive 5 Epistemic Restraint, rate-limit backoff, and offline deterministic fallback.*
Stage 7: Citation Normalization & Media Verifier • *Converts shorthand to [Doc, Page X], verifies image disk paths, and automatically injects visual plates if omitted by LLM.*
Stage 8: Interactive Multimodal Presentation (Streamlit) • *Renders formatted answers, inline images, data tables, and expandable multi-chunk telemetry.*

Multimodal Ingestion & Chunking Contract (chunks.json)

Ingestion processes 415 mixed-format documents into a unified schema contract: chunk_id, document_name, page_number, modality (text, table, image-caption), content, media_path, and metadata. Tables are parsed with layout-aware boundary heuristics and serialized to Markdown tables. Figures and plate diagrams are cropped at full resolution into data/extracted_media/figures/ with verified disk links.

4. Key Technical Decisions & Engineering Rationale

Decision Area	Chosen Approach	Alternatives Considered & Why Rejected
Embedding Model	**Local BAAI/bge-small-en-v1.5** Zero cost, 100% offline, persistent ChromaDB.	Voyage AI / OpenAI API: Exposes external network latency and rate-limit risks during multi-hop benchmarks.
LLM Inference Engine	**Groq Cloud LPU (openai/gpt-oss-120b)** 300+ tokens/sec, ~1.35s latency, free tier.	Local 70B LLM (untenable RAM requirements on laptop); Commercial frontier APIs (violated zero-cost principle).
Context Sizing	**Autonomous 3-Stage Budgeting** Intent K sizing (2..8) + Elbow drop-off (0.60).	Fixed K=5 (distractor noise on facts; missed secondary sources on matrices); Manual UI sliders (bad UX).
Contradiction Handling	**Strict Epistemic Restraint** Verbatim reporting of conflicting sources.	Speculative narrative bridging: LLM hallucinated ungrounded motives for archival omissions.

5. What Works, What Doesn't, & System Limitations

In accordance with the competition rubric's emphasis on **Honest Engineering Evaluation (10%)**, we report an objective accounting of verified capabilities, known boundaries, and failed early approaches.

5.1 What Works (Demonstrated Capabilities)

- **Direct Multimodal Inlining:** Seamlessly extracts, references, and renders figure plates, diagrams, and markdown tables directly inside generated answers.
- **Zero-Hallucination Epistemic Guarding:** Reports conflicting accounts (e.g. Imperial Codex vs Tavern Song) verbatim without fabricating reconciling lore.
- **Pre-Generation Scope & Premise Gate:** Refuses out-of-scope inquiries (real-world people, math, coding, physical sensor queries) instantly (<0.005s) with 3 helpful archival options.
- **Citation Normalization:** Automatically normalizes shorthand citations (e.g. [Chunk 1]) into verified document names and page numbers.
- **Deterministic Grounded Fallback:** If API quotas or network connections drop, the offline deterministic synthesizer generates grounded answers with valid images.

5.2 Known Limitations & Edge Cases

- **Degraded Ephemera Scans:** Simulated scanned letters with heavy blur or handwritten marginalia can occasionally yield partial OCR character errors under standard Tesseract.
- **Deeply Nested Multi-Header Tables:** Highly irregular codex tables with merged 3-tier headers can lose column boundaries when flattened to Markdown. *Mitigation: DeepThink embeds the original visual plate alongside the table.*
- **Groq Free-Tier Rate Limits:** Heavy back-to-back testing can trigger HTTP 429 TPM limits. DeepThink mitigates this via dynamic error string wait parsing and auto-fallback to high-capacity models (llama-3.3-70b-versatile).

5.3 Approaches Tried and Discarded (Lessons Learned)

1. Discarded: Naive Flat Text Chunking (Phase 1 Baseline)
What happened: Stripped all media tags and treated documents as flat 500-token text chunks. Resulted in 90% failure on visual questions: the LLM hallucinated visual descriptions without embedding actual plates, and table columns were scrambled.

2. Discarded: Full Vision-LLM on Every Query
What happened: Feeding raw PDF page image screenshots to a large Multimodal Vision LLM exhausted free token quotas within 5 queries and caused 8–12 second latencies. Discarded in favor of *Modality-Aware Hybrid Routing*.

3. Discarded: Static K=5 Retrieval with Manual UI Sliders
What happened: Fixed K=5 introduced 3-4 distractor chunks for simple fact questions and missed secondary sources for comparative matrices. Replaced with autonomous 3-stage adaptive budgeting.

4. Discarded: Speculative Narrative Bridging
What happened: Prompting the LLM to reconcile contradictory records caused it to fabricate motives like 'divergent narrative framing'. Replaced with strict epistemic restraint (verbatim conflicting citation).

6. Evaluation Results & Benchmark Suite (sample_questions.json)

We tracked system performance across 3 distinct engineering phases against the official 20 development questions in sample_questions.json:

Metric	Phase 1 (Baseline)	Phase 2 (Modality-Aware)	Phase 3 (Compound RAG 3.0)	Target
Retrieval Precision (Top-5)	60.0% (12/20)	85.0% (17/20)	**100.0% (20/20)**	≥ 90.0%
Citation Accuracy	55.0% (11/20)	90.0% (18/20)	**100.0% (20/20)**	≥ 95.0%
Hallucination Rate	20.0% (4/20)	5.0% (1/20)	**0.0% (0/20)**	≤ 5.0%
Modality Success (Figures/Tables)	10.0% (1/10)	90.0% (9/10)	**100.0% (11/11)**	≥ 90.0%
End-to-End Latency	2.1s (no images)	2.8s	**1.35s (Groq LPU)**	< 3.5s

7. AI Usage Disclosure (Section 4.1 Compliance)

In accordance with **Section 4.1 (AI Usage Policy)**, DeepThink provides full, transparent disclosure regarding the use of AI tools. AI coding assistants were leveraged strictly as accelerators and intellectual sounding boards, while all engineering decisions, core designs, and quality verification were steered by human team members.

AI Tool / Model	Primary Role in Project	Human Oversight & Control
Claude 3.7 Sonnet / Google Antigravity	Brainstorming architectural patterns, drafting parser boilerplate, syntax corrections, and refining prompt directives.	All code locally reviewed, executed, and adapted against real corpus PDFs. Directives audited to eliminate hallucinated narrative framing.
Groq Cloud (openai/gpt-oss-120b)	Runtime inference engine for grounded context synthesis, citation substantiation, and markdown plate embedding.	Constrained by strict epistemic prompts (Directive 5). Monitored for rate-limit backoff and backed by deterministic offline fallback.
HuggingFace (BAAI/bge-small-en-v1.5)	Generating dense vector representations of chunks for local ChromaDB indexing and retrieval.	Selected by team to replace cloud embedding dependencies, ensuring 100% zero-cost reproducibility on judges' machines.

Human vs. AI Decision Boundary:

- **Conceived Exclusively by Human Team:** Sub-track 1A focus, modality-tagged chunks.json contract, 3-stage adaptive budgeting ($K=2..8$ + Elbow cutoff), character dossier boost (+0.35), multi-entity intersection (+0.40), and pre-retrieval categorized fast-path guardrail.
- **Assisted by AI:** Fast drafting of PyMuPDF extraction scripts, regex citation matching patterns, Streamlit layout scaffolding, and unit test boilerplate.
- **Exported Chat Logs:** Complete, unedited plain-text logs of all AI development sessions are exported to ai_usage/claude.txt and ai_usage/claude.md.

8. Team Member Names, Roles, & Contributions

The DeepThink engineering team consists of four undergraduate members with clearly defined ownership across the modular system architecture:

Member Name	Role & Module Ownership	Key Technical Contributions
Fatheen M. F. A.	**P1 Lead** Document Parsing & OCR	Built ingestion pipelines across 415 documents (PDF, DOCX, MD, TXT); integrated OCR for degraded simulated scans; maintained section hierarchy.
M. M. M. Shakeer	**P2 Lead** Visual & Table Extraction	Extracted 86+ visual plates, diagrams, and maps; cropped high-res figures to disk; converted codex data tables to structured Markdown; authored plate metadata.
Rasheed A. A. A.	**P3 Lead / Team Lead** Retrieval & Ranking	Engineered Compound RAG 3.0; local ChromaDB indexing with BGE embeddings; 3-stage adaptive context budgeting; agentic query decomposition; re-ranking.
S. Dharshan	**P4 Lead** Generation, Evaluation & UI	Groq Cloud LPU integration; strict epistemic prompting; automated 20-question benchmark runner (evaluator.py); Streamlit UI; report and demo video.

9. Reproducibility, Verification & Engineering Best Practices

A critical criterion in the competition rubric is **Engineering Best Practices (15%)**: *'Can a judge set up and run your project from the README alone?'* DeepThink is architected for zero-friction judge verification.

9.1 Four-Step Reproducibility Protocol

```
# 1. Clone repo & create virtual environment
git clone <repo_url> && cd DeepThink
python3 -m venv venv && source venv/bin/activate

# 2. Install dependencies & configure free Groq key
pip install -r requirements.txt
cp configuration-example/.env.example .env

# 3. Run automated test suite (75 tests)
python -m unittest discover -s tests

# 4. Launch interactive Multimodal Streamlit Assistant
streamlit run src/ui/app.py
```

9.2 Automated Test & Verification Coverage

The repository contains an exhaustive test suite of **75 automated unit and integration tests** passing cleanly with 0 errors:

- tests/test_boundary_guardrails.py: Validates out-of-scope refusals, slurs, sensor queries, and fast-path execution.
- tests/test_rag3.py: Tests adaptive budget calculation (K=2..8), agentic decomposition, and quantitative table aggregation.
- tests/test_retrieval_verification.py: Tests Elbow-method drop-off pruning (score < 0.60 × TopScore) and pre-gen entity gating.
- tests/test_generator.py: Tests citation normalization ([Chunk 1] → [Doc, Page]), media path verification, and grounded fallbacks.
- tests/test_benchmark_evaluation.py: Executes end-to-end smoke evaluations verifying precision, citations, and image embeds.

10. Conclusion & Final Round Defense Readiness

DeepThink represents a rigorous, production-grade response to **SLIIT Codefest 2026 Sub-track 1A**. By treating visual assets and tables as first-class evidentiary citizens rather than secondary text descriptions, DeepThink solves the central failure mode of enterprise document assistants. The system combines frugal engineering (100% free local embeddings and zero credit card requirements) with hardware-accelerated Groq LPU inference to deliver rich, verifiable, multimodal answers in under 1.5 seconds. Every member of the DeepThink team has mastered their subsystem and is prepared to demonstrate live execution, modify code on demand, and defend our architectural choices before the judging panel at the SLIIT Main Auditorium.

Team DeepThink: *Fatheen M. F. A. • M. M. M. Shakeer • Rasheed A. A. A. • S. Dharshan*
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